

Data management and use: case studies of technologies and governance

Produced for the British Academy and the Royal Society

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2. Introduction

These case studies were prepared in support of the *Data Management and Use: Governance in the 21st Century* project, carried out by the British Academy and the Royal Society. The purpose of these case studies was to stimulate thinking and discussion by the working group, in their deliberations on governance needs for data management and use across all sectors. This publication is an edited version of the case studies presented to the working group.

The case studies aim to give concrete examples of the kinds of social and ethical tensions that arise in contemporary data use and management – and they draw on the sets of social and ethical pairings, detailed in the main report of this project, that were identified at a cross-disciplinary expert workshop held in July 2016. They give current and forward-looking examples of the benefits and challenges of data collection, management and use across a range of sectors and the governance needs in different contexts.

Intended as they are solely to inform the working group, these case studies are not presented as the views of either Academy, and are not intended as appraisals or evaluations of any of the governance approaches identified in them. However they illustrate some of the issues that prompted the work behind the *Data Management and Use: Governance in the 21st Century* report.

The case studies were developed using desk research and informal interviews with researchers. Each case study has been reviewed by a relevant expert, as listed in the acknowledgements above.

3. Detailed Case Study: Smart Metering

3.1 Introduction

Smart meters are devices capable of real-time measurement and transmission of household electricity consumption. The UK Government, similar to many others around the world, aims to 'ensure that every home and business in the country is offered a smart meter by 2020, delivered as cost effectively as possible'. Rollout is non-compulsory, led by BEIS¹, regulated by Ofgem, and primarily funded and fulfilled by the energy suppliers. They promise to provide new opportunities for innovative markets, efficiencies in transmission, maintenance and billing, and spillover effects concerning 'smart homes' more generally. At the same time, they raise concerns around privacy, proportionality and security of energy systems.

This extended case study sets out the social and ethical tensions that are relevant to the management and use of (primarily domestic) energy data, from a selection discussed in greater length in the main report, *Data Management and Use: Governance in the 21st Century*. It considers the opportunities and challenges in using smart meter data, and looks at the ways that data management and use can be governed through technology and institutions. Finally it gives an overview of the governance arrangements in place in the UK and Europe. The case study also illustrates the challenge for governments in deploying cutting edge, technical data governance solutions where there is a one-off national roll-out of an infrastructure system.

3.2 Social and ethical issues in smart meter data management and use

Protecting personal information while safely using and linking open and non-sensitive data

The individual accounting of data at household level is the source of many of the attributed benefits of smart metering, yet also serves as the source of many of the identified risks. This individual accounting is essential for accurate billing, for example. However, live electricity consumption data can enable the inference of private data such as when you are at home to what you do when you are there.

Proportionality in the use of data while using data to protect public safety and wellbeing

The detailed inferences possible with smart metering data might provide opportunities for social uses, such as targeting services to vulnerable individuals, or building more detailed

¹ The rollout was formerly led by the Department for Energy and Climate Change (DECC) before departmental reorganisation in 2016 shifted the portfolio to the new Department of Business, Energy and Industrial Strategy.

maps of deprivation. However, consumers may be wary of such data collection, and maintaining public trust could prove difficult where there are demands for data repurposing.

Autonomy for individuals and communities while using data to achieve commercial benefit and efficiency in public services

In the UK, a single, centralised model for smart meter data governance was promoted by the regulatory approach, to create a specific new licensed body (a Data and Communications Company) for data processing in domestic energy regulation. To attempt to manage the privacy issues within this model, consumers were provided with rights over the granularity of data they send. However, this creates a challenge as European cost-benefit analyses demonstrate that the systems are only broadly cost effective to install if operational benefits are largely realised, based on detailed data being available. As discussed later in this example, the German system illustrates a way of addressing this tension.

3.3 Opportunities for use of smart meter data

Smart meters present opportunities for both energy efficiency, through incentivising the consumer and better grid management, and for future business models, such as the charging of electric vehicles away from home. Previous studies have distinguished between benefits emerging from better management of aggregated operations and benefits which emerge from more flexible billing². A third category can be added: spillover benefits that might emerge from the repurposing of the metering infrastructure. Considering these opportunities together has been an important part of the calculus for smart meter installation. At a European level, the cost of smart meter installation is not fully offset by operational savings, requiring dynamic pricing provisions to make the present value of benefits overtake the costs³.

Operational opportunities can result in efficiency savings for energy networks and providers. Data on electrical usage has previously only been routinely possible to obtain at a substation level at useable temporal resolutions. Smart meters providing data at higher resolution are expected to provide a range of benefits to suppliers and other energy decision-makers. These benefits include the better *projection of future network capacity requirements*; enabling *preventative maintenance*; and aiding in *understanding of faults, outages and quality issues*⁴.

Billing opportunities allow for the creation of new markets, and are promoted as enhancing choice and efficiency. Capture of electricity usage information at a higher temporal resolution creates opportunities for *new tariff models*. Demand responsive pricing might serve to lower consumption at peak times, as well as incentivise action on energy wastage. Smart meters might also *incentivise consumer energy generation* through responsive feed-in tariffs⁵, and

² Jawurek M, Kerschbaum F, George D. 2012 SoK: Privacy technologies for smart grids – A survey of options. *Microsoft Research*. See <https://www.microsoft.com/en-us/research/wp-content/uploads/2012/11/paper.pdf> (accessed 26 June 2017); Finster S, Baumgart I. 2015 Privacy-aware smart metering: A survey. *IEEE Communications Surveys Tutorials*. **17**, 1088-1101. (doi:<http://dx.doi.org/10.1109/COMST.2015.2425958>)

³ Faruqi A, Harris D, Hledik R. 2010 Unlocking the €53 billion savings from smart meters in the EU: How increasing the adoption of dynamic tariffs could make or break the EU's smart grid investment. *Energy Policy*. **38**, 6222-6231. (doi: <https://doi.org/10.1016/j.enpol.2010.06.010>)

⁴ Depuru SSSR, Wang L, Devabhaktuni V, Gudi N. 2011 Smart meters for power grid: Challenges, issues, advantages and status. *Renewable and Sustainable Energy Reviews*. **15**. (doi: <http://dx.doi.org/10.1109/PSCE.2011.5772451>)

⁵ Römer B, Reichhart P, Kranz J, Picot A. 2012 The role of smart metering and decentralized electricity storage for smart grids: The importance of positive externalities. *Energy Policy*. **50**, 486-495. (doi: <https://doi.org/10.1016/j.enpol.2012.07.047>)

provide *cost savings using flexible energy storage devices*⁶. The ability to issue bills in a more timely fashion can strengthen this incentive link, in addition to reducing labour costs in call-centres and meter reading.

Smart meters also increase the capabilities of suppliers to *detect and act on fraud and non-payment*. Fraud detection can cost suppliers greatly, and increase electricity costs in general. British Gas reports that electricity theft costs the industry £400–500m a year, while worldwide energy theft equates to the total installed generation capacity of the UK, Germany and France together⁷. Even partial prevention might significantly reduce carbon emissions⁸.

Some smart meters have the capability to be disabled remotely, which might be used to deter fraud and non-payment. Remote management of meters can aid in the management of pre-pay meters online and allow *accurate billing of costs* while away. It also allows for new business models, particularly around flexible charging of electric cars while away from home⁹.

Infrastructural benefits have been highlighted in relation to the installation of these technologies. Smart meters allow the better *visualisation of energy consumption*, which might serve to reduce consumption¹⁰— although this can be accomplished ‘offline’ without transmission beyond the home¹¹. Smart meters have also been proposed as *hubs for ‘Internet of Things’ technologies*, due in part to their wide public rollout. These hubs present both opportunities and risks, as discussed below.

3.4 Challenges in the management and use of smart meter data

In addition to these opportunities, smart meters also present certain privacy and security hurdles.

Privacy concerns stem from the fact that *real-time data from smart meters can leak information a user might want to remain private*. There are several categories of information possible to infer from smart meter readings:¹²

- Which appliances are being used at particular times¹³

⁶ Malhotra A, Battke B, Beuse M, Stephan A, Schmidt T. 2016 Use cases for stationary battery technologies: A review of the literature and existing projects. *Renewable and Sustainable Energy Reviews*. **56**, 705–721. (doi:<https://doi.org/10.1016/j.rser.2015.11.085>); Stephan A, Battke B, Beuse MD, Clausdeinken JH, Schmidt TS. 2015 Limiting the public cost of stationary battery deployment by combining applications. *Nature Energy*. **1**. (doi: <http://dx.doi.org/10.1038/nenergy.2016.79>)

⁷ Depuru SSSR, Wang L, Devabhaktuni V. 2011 Electricity theft: Overview, issues, prevention and a smart meter based approach to control theft. *Energy Policy*. **39**, 1007–1015. (doi:<https://doi.org/10.1016/j.enpol.2010.11.037>)

⁸ Pyasi A, Verma V. 2008 Improvement in electricity distribution efficiency to mitigate pollution IEEE ISEE. *IEEE International Symposium on Electronics and the Environment*. (doi: dx.doi.org/10.1109/ISEE.2008.4562863)

⁹ Clement-Nyons K, Haesen E, Driesen J. 2010 The impact of charging plug-in hybrid electric vehicles on a residential distribution grid. *IEEE Transactions on Power Systems*. **25**, 371–380. (doi: <http://dx.doi.org/10.1109/TPWRS.2009.2036481>)

¹⁰ Fischer C. 2008 Feedback on household electricity consumption: A tool for saving energy? *Energy Efficiency*. **1**, 79–104. (doi: <http://dx.doi.org/10.1007/s12053-008-9009-7>)

¹¹ To note, some studies cast doubt on the ability of current technologies to incentivise lower energy consumption through visualisation alone. Buchanan K, Russo R, Anderson B. 2015 The question of energy reduction: The problem(s) with feedback. *Energy policy*. **77**, 89–96. (doi: <https://doi.org/10.1016/j.enpol.2014.12.008>)

¹² Jawurek M, Kerschbaum F, George D. 2012 SoK: Privacy technologies for smart grids – A survey of options. *Microsoft Research*. See <https://www.microsoft.com/en-us/research/wp-content/uploads/2012/11/paper.pdf> (accessed 26 June 2017).

¹³ Molina-Markham A, Shenoy P, Fu K, Cecchet E, Irwin D. 2010 Private memoirs of a smart meter. *Proceedings of the 2nd ACM Workshop on Embedded Sensing Systems for Energy-Efficiency in Building*. See <http://dl.acm.org/citation.cfm?doid=1878431.1878446> (accessed 2 November 2016).

- How these appliances are used. Attacks by researchers have been able to detect particular TV channels or websites visited from electricity data¹⁴
- Behavioural patterns deduced from patterns of appliance use¹⁵
- Inferred information about other utilities or features of the building¹⁶
- Information that can be used to identify other pseudonymised records¹⁷.

This information can lead to the inference of private knowledge. This knowledge might include times you tend to be home; length and frequency of your showers; whether you are protected by an alarm; how often you are out late; whether you tend to leave appliances on¹⁸.

Security concerns can stem from remote shutdown, which can present a risk to national infrastructure if meters are insecure – allowing attacks to be targeted at times of peak demand, and potentially leading to grid damage^{19,20}. Understanding security threats to smart grids requires a cyber-physical framing, where both cyber attacks and physical attacks (and combinations of both) have both cyber and physical consequences²¹. There is also a risk of unauthorised third parties obtaining data flows. These concerns could be compounded by the potential for smart meters serve as ‘hubs’ for connected domestic ‘IoT’ devices, making them attractive targets for cybercriminals. Significant concerns have already been raised around the security protocols in meter-reading devices in the US²². Risks are exacerbated where homogenous hardware could create systemic vulnerabilities.

Proportionality and consent are issues in accessing smart meter data. There is a trade-off between giving detailed information that might be important to some users, and ensuring that consumers can understand the information presented to them to give truly informed consent (known as the ‘transparency paradox’)²³. Secondly, tariffs using more granular data are likely

¹⁴ Enev M, Upgta S, Kohno T, Patenl SN. 2011 Televisions, video privacy, and powerline electromagnetic interference. *Proceedings of the 18th ACM Conference on Computer and Communications Security*. (doi:<http://doi.acm.org/10.1145/2046707.2046770>); Greveler U, Glösekötter P, Justus B, Loehr D. 2012 Multimedia content identification through smart meter power usage profiles. *Proceedings of the International Conference on Information and Knowledge Engineering (IKE)* (The Steering Committee of The World Congress in Computer Science, Computer Engineering and Applied Computing (WorldComp) 2012). See <http://search.proquest.com/openview/059b0c797d4580cd7419dc90a2462602/1?pq-origsite=gscholar> (accessed 26 June 2017); Clark SS, Ransford B, Sorber J, Xu W, Learned-Miller E, Fu K. 2013 Current events: Identifying webpages by tapping the electrical outlet. In Crampton J, Jajodia S, Mayes K (eds), *Computer Security – ESORICS 2013* (Springer Berlin Heidelberg 2013). See http://link.springer.com/chapter/10.1007/978-3-642-40203-6_39 (accessed 2 November 2016).

¹⁵ Lisovich MA, Mulligan DK, Wicker SB. 2010 Inferring personal information from demand-response systems. *IEEE Security Privacy*. 8, 11. (doi:<http://dx.doi.org/10.1109/MSP.2010.40>); Beckel C, Sadamori L, Thorsten S, Silvia S. 2014 Revealing household characteristics from smart meter data. *Energy*. 78, 397-410. (doi:<http://dx.doi.org/10.1016/j.energy.2014.10.025>)

¹⁶ Beckel C, Sadamori L, Thorsten S, Silvia S. 2014 Revealing household characteristics from smart meter data. *Energy*. 78, 397-410. (doi:<http://dx.doi.org/10.1016/j.energy.2014.10.025>)

¹⁷ Jawurek M, Johns M, Rieck K. 2011 Smart metering de-pseudonymization. *Proceedings of the 27th Annual Computer Security Applications Conference*. (doi:<http://doi.acm.org/10.1145/2076732.2076764>); Tudor V, Almgren M, Papatriantafillou M. 2015 A study on data de-pseudonymization in the smart grid. *Proceedings of the Eighth European Workshop on System Security*. (doi:<http://dl.acm.org/citation.cfm?doid=2751323.2751325>)

¹⁸ Quinn EL. 2009 Privacy and the New Energy Infrastructure. (doi:<http://dx.doi.org/10.2139/ssrn.1370731>)

¹⁹ Anderson R, Fuloria S. 2010 Who Controls the off Switch? *First IEEE International Conference on Smart Grid Communications*. (doi:<http://dx.doi.org/10.1109/SMARTGRID.2010.5622026>)

²⁰ At a major 2014 security conference a flaw in Spanish smart meters was demonstrated which could theoretically result in an attacker being able to turn off the electricity in a wide region, potentially also causing significant grid damage. Illera AG, Vidal JV. 2014 Lights off! The darkness of the smart meters. *BlackHat Europe*. See https://www.youtube.com/watch?v=Z_y_vjYtAWM (accessed 26 June 2017).

²¹ Mo Y et al. 2012 Cyber-physical security of a smart grid infrastructure. *Proceedings of the IEEE*. 100, 195-209. (doi:<http://dx.doi.org/10.1109/JPROC.2011.2161428>).

²² Rouf I, Mustafa H, Xu M, Xu W, Miller R, Gruteser M. 2012 Neighborhood watch: Security and privacy analysis of automatic meter reading systems. *Proceedings of the 2012 ACM conference on Computer and communications security*. (doi:<http://dx.doi.org/10.1145/2382196.2382246>).

²³ Nissenbaum H. 2011 A contextual approach to privacy online. *Daedalus*. 140, 32-48. (doi:http://dx.doi.org/10.1162/DAED_a_00113)

to be cheaper, meaning that lower income households might be pushed to consent to greater data sharing for financial reasons.

Several technical solutions for these issues have been proposed, many using cutting-edge research in privacy-enhancing technologies to maximise the aggregate benefits of smart meters while not revealing individuals' individual consumption patterns. Yet these technologies have not been utilised in emerging governance systems, which differ between countries. The UK has chosen to centralise all data processing and access through a single new company while Germany has created a market for a new certified category of device, the smart meter 'gateway', which allows personalised end-to-end encryption between the home and various data users.

3.5 Governance of smart meter data management and use

Governance of smart meters to date can be seen through two interlinked lenses of **technical governance** and **institutional governance**.

Technical governance: Privacy-enhancing technologies

Privacy issues relating to smart meters have been the subject of many proposed **privacy-enhancing technologies (PETs)**, which can be seen as a form of technical governance. PETs are

*a system of ICT measures protecting informational privacy by eliminating or minimising personal data thereby preventing unnecessary or unwanted processing of personal data, without the loss of the functionality of the information system.*²⁴

A range of PETs are applicable to both the operational and the billing functions of smart meters²⁵.

Operationally-relevant PETs attempt to maintain user informational privacy while allowing suppliers and other organisations to benefit from the rich information and functionality that these systems provide – potentially addressing the tension between protecting personal information and making use of non-sensitive data. A core privacy aim is to allow aggregate summary statistics of multiple readings without revealing information from individual meters. *Anonymisation* tools seek to remove the links between data creators and data users in ways that still allow desired calculations to be carried out. However, anonymisation tools can suffer from re-identification attacks and there is a trade-offs between more thorough anonymisation techniques and the accuracy of statistics generated from this data²⁶. *Perturbation* seeks to

²⁴ Several wide-ranging surveys of these systems exist, see: Van Blarckom GE, Borking JJ, Olk J. 2003 Handbook of privacy and privacy-enhancing technologies. *Privacy Incorporated Software Agent (PISA) Consortium, The Hague*. See http://www.andrewpatrick.ca/pisa/handbook/Handbook_Privacy_and_PET_final.pdf (accessed 26 June 2017).

²⁵ Jawurek M, Kerschbaum F, George D. 2012 SoK: Privacy technologies for smart grids – A survey of options. *Microsoft Research*. See <https://www.microsoft.com/en-us/research/wp-content/uploads/2012/11/paper.pdf> (accessed 26 June 2017); Finster S, Baumgart I. 2015 Privacy-aware smart metering: A survey. *IEEE Communications Surveys Tutorials*. 17, 1088-1101. (doi:<http://dx.doi.org/10.1109/COMST.2015.2425958>)

²⁶ Daries J et al. 2014 Privacy, anonymity, and big data in the social sciences. *Communications of the ACM*. 56. See <http://dl.acm.org/citation.cfm?doid=2663191.2643132> (accessed 3 November 2016); Angiuli O, Waldo J. 2016 Statistical tradeoffs between generalization and suppression in the de-identification of large-scale data set. *IEEE 40th Annual Computer Software and Applications Conference (COMPSAC)*. (doi:<http://dx.doi.org/10.1109/COMPSAC.2016.198>)

introduce noise into readings that preserves some privacy while maintaining the utility of the resultant calculations. *Cryptographic computation* encrypts data using a homomorphic protocol²⁷ which allows mathematical operations to be carried out while data is still encrypted, while allowing the decryption of the final result but not any of the individual parts.

PETs designed for billing require the waiving of anonymity, and the guarantee of accuracy, in order to correctly charge customers, and to enable dynamic pricing and tariffs. Billing however does not have to be reported in real time, but can be sent in batches at a coarse enough temporal resolution as to suppress most privacy concerns. This, however, can cause issues with the desire of suppliers to use dynamic pricing and tariffs. One technical solution to this is *secure multi-party computation*, which assumes the existence of one or more third parties external to the data user. The data user receives billing information resulting from other parties jointly computing a function over split inputs while keeping those inputs private to each other. A different solution, and one which avoids the use of third parties, is *verifiable computing*. Here the data creator themselves provides a record alongside the aggregate consumed which mathematically proves a computation was carried out in a particular way, while providing zero knowledge beyond the veracity of the calculation.

Institutional governance

There are a range of institutional governance needs in relation to data collection; transmission; processing, storage and access; and interoperability. This section sets out some of the responses to those needs.

Data collection

The Netherlands was an early actor in smart meter governance²⁸. Initial legislative plans developed in 2006 focussed on energy efficiency, and mandated citizens to install a smart meter that would transfer data every fifteen minutes and was enabled with remote shutdown functionality. Failure to install would be met with a 17,000 EUR fine or even maximum of six months imprisonment. A report commissioned by the Dutch Consumer organisation, Consumentenbond, argued that the proposed rollout would violate Article 8 of the European Convention on Human Rights, the respect for private and family life²⁹. This led to new legislation being proposed giving Dutch consumers both options to refuse smart meters with administrative shutdown capabilities or the ability to read a continuous stream of data, and to reject the installation of a smart meter entirely.

The Dutch case highlighted the phenomenon of ‘function creep’ with regards to data governance. The initial European-level proposals which led to the introduction of the smart meter legislation specified only the use of smart meters for providing information to end-users to help them save energy. The Dutch implementation of such legislation added abilities to combat electricity fraud and remote activation and shutdown, which were two of the features that caused public outrage.

²⁷ Gentry C. 2009 A fully homomorphic encryption scheme. *Stanford University*. See <http://cs.au.dk/~stm/local-cache/gentry-thesis.pdf> (accessed 26 June 2017).

²⁸ Cuijpers C, Koops B. 2013 Smart metering and privacy in Europe: Lessons from the Dutch case. In Gutwirth S *et al* (eds), *European Data Protection: Coming of Age* (Springer Netherlands 2013). See http://link.springer.com/chapter/10.1007/978-94-007-5170-5_12 (accessed 3 November 2016).

²⁹ Cuijpers C, Koops B. 2008 Het Wetsvoorstel “slimme Meters”: Een Privacytoets Op Basis van Art. Universiteit van Tilburg. See https://www.vrijbit.nl/images/stories/files/pdf/onderzoek_uvt_slimme_energi1.pdf (accessed 26 June 2017).

In the UK, the then Department of Energy and Climate Change (DECC) began in 2010 to discuss privacy in relation to their own plans for smart meter rollout³⁰. The UK consumer group, Consumer Focus, using their statutory powers found that trials of smart readers were already collecting half-hourly data readings.

While DECC was drawing up final plans, with reference to a call for evidence and a consultation, the Article 29 Working Party offered an opinion³¹ suggesting that data from smart meter operations was personal data, and therefore data protection legislation applied to it. In the same opinion, they warned that the creation of detailed profiles might not be in line with data subjects' legitimate interests, and was not needed to achieve the basic purposes of smart metering.

Technical governance mechanisms, such as those described above, have not found great traction in the law, although there is general support for them in both the Data Protection Directive and the General Data Protection Regulation which takes force in Europe in 2018. For measurement relating to billing, metrology law³² covers device accuracy, an important component of the governance of data collection and sensor validity.

Data transmission

The transmissions techniques used by smart meters vary widely³³. The cheapest is power-line communication (PLC), which uses existing electrical cables to simultaneously carry data. PLC requires less fixed investment than other options, yet can provide lower signal qualities and data speeds³⁴, which might serve to limit the type of technical governance possible, particularly for types of trusted computing that require significant network interaction. Other options proposed have included SMS over GSM, local wireless and bluetooth networks, mobile 3G and 4G telecoms, and standard telephone lines. Where the meters use broader infrastructure, part of the governance of data during its transmission, including issues such as its security, falls under existing governance systems.

In the UK case, a mix of existing infrastructure and new infrastructure will be used to connect smart meters to a central data services provider (DSP) that will distribute data to suppliers, distribution network operators (DNOs) and authorised third parties. In the south and central parts of the UK, this infrastructure will be provided by Telefónica using existing cellular networks combined with a wireless mesh network in low-coverage areas, while Arqiva will build new infrastructure in the north.

Before the rollout of smart meters with the national data management system in the UK, many suppliers already installed 'remote access meters' with similar functionality. In 2015, Ofgem decided to extend the privacy requirements for smart meters to remote access meters, following a consultation³⁵, yet these plans faced hardware challenges, as many of the smart meters already installed were unable to be reconfigured to the consumer's wishes. Ofgem,

³⁰ Brown I. 2014 Britain's smart meter programme: A case study in privacy by design. *International Review of Law, Computers & Technology*. **28**, 172-184. (doi:<http://dx.doi.org/10.1080/13600869.2013.801580>)

³¹ 00671/11/EN WP 183 Article 29 Data Protection Working Party: Opinion 12/2011 on smart metering.

³² Of particular relevance is the Measuring Instruments Directive (2004/22/EC).

³³ Khalifa T, Naik K, Nayak A. 2010 A survey of communication protocols for automatic meter reading applications. *IEEE Communications Surveys Tutorials*. **13**, 168-182. (doi:<http://dx.doi.org/10.1109/SURV.2011.041110.00058>)

³⁴ Gungor VC et al. 2012 Smart grid and smart homes: Key players and pilot projects. *IEEE Industrial Electronics Magazine*. **6**, 18-34. (doi:<http://dx.doi.org/10.1109/MIE.2012.2207489>)

³⁵ Ofgem. 2015 Decision on extending the smart meter framework for data access and privacy to remote access meters. See <https://www.ofgem.gov.uk/ofgem-publications/93187/dataprivacyextension-decision1.pdf> (accessed 26 June 2017).

with the support of the Information Commissioner's Office (ICO), proposed that for those systems, the companies transmitting from the remote meters to the suppliers (so-called 'head end operators') are compelled by the regulator to filter user data in alignment with the given consent.

Data processing, storage and access

Different models of data processing and access exist in relation to smart meters. This section summarises three different implementations, from the UK, the Netherlands and Germany.

In **the UK**, data from smart meters is fed to a centralised authority, known as the Data and Communications Company (DCC)³⁶, envisioned as a central point for data to be provided to suppliers, operators of distribution systems, as well as authorised third parties³⁷. The processes behind authorisation to use this data are split between government and regulators, which will oversee the processes underlying access to these data streams; and consumers themselves, required to consent to the making available of any data beyond that used for billing – though customers will not have to consent where data are 'required to fulfil regulated duties'³⁸. In addition, while meters will be installed with some simple variable tariffs, such as algorithms based on common times of day, consent to transmission of more granular data will be necessary for consumers to access more advanced tariffs that are likely to save them money. No technical privacy-enhancing technologies are being publicly deployed within the data accessible to the DCC³⁹, and it was reported that DECC saw these technologies as yet immature for use, and industry argued that designers of privacy-enhancing technologies 'didn't understand the industry'⁴⁰.

In **the Netherlands**, the data from smart meters is fed into a central system, which often operates at a substation level. From this level, it flows to the operator of the distribution system and to energy suppliers. In theory, pre-processing *could* happen at this neighbourhood level, such as aggregation, where the supplier and other parties need only know that the substation was a trusted processor, rather than access consumers' detailed data. No technical governance of this sort has yet been published alongside the broader system design, however.

Germany presents a contrasting case to both the UK and the Dutch approaches. Firstly Germany affords the consumer the right to switch to a third party organisation to operate their smart meter and to collect their data. Secondly, as opposed to a system with data 'funnels' such as the DCC, data is centralised in consumers' homes through the use of a 'gateway' device. This is a device certified by the regulator, and provided by a consumer-chosen 'measuring point operator' which links to one or more smart meters.⁴¹ The underlying data transfer principle is one of 'star-shaped communication', where this gateway can communicate with different protocols with the different users of smart meter data and third parties on the

³⁶ This organisational role was put out to tender and granted to Smart DCC Limited, a subsidiary of Capita plc.

³⁷ For example, the suggestion has been made that Citizens Advice could use such data to provide targeted services to vulnerable individuals: House of Commons Science and Technology Committee, *Evidence Check: Smart Metering of Electricity and Gas* (House of Commons 2016).

³⁸ They can however access at least the previous 13 months of data locally on their own meter. See Connor PM *et al.* 2014 Policy and regulation for smart grids in the United Kingdom. *Renewable Sustainable Energy Rev.* **40**, 269-286. (doi:<http://dx.doi.org/10.1016/j.rser.2014.07.065>)

³⁹ Ibid.

⁴⁰ Brown I. 2014 Britain's smart meter programme: A case study in privacy by design. *International Review of Law, Computers and Tech.* **28**, 172-184. (doi:<http://dx.doi.org/10.1080/13600869.2013.801580>)

⁴¹ See the Gesetz zur Digitalisierung der Energiewende [trans: Law for the Digitisation of the Energy Transition] (adopted August 29 2016).

basis of consumer authorisation. This implies end-to-end encrypted communication with no middle data processing body⁴².

It is important to note that products that somewhat resemble smart meters, and interact with home appliances, such as smart thermostats, may interact with smart technologies. By the same logic of the Article 29 Working Party in relation to government smart meter roll out, this data is also personal data, and therefore is similarly covered by data protection legislation. It does not, however, benefit from any of the extra rules that national governments have placed on data use and consent in specific relation to smart meter data, and consumers may not be aware that two or more transmission and processing activities are happening, often using two distinct transmission modalities.

Data interoperability

Meters adopted in the UK prior to mass roll-out pose significant barriers to interoperability. Data collection systems and transmission systems would not necessarily be able to work in 'smart' mode when supplier was switched. DECC noted that interoperability of these systems and the transfer of data-driven services was 'subject to agreement between energy suppliers', noting that the industry trade body Energy UK was 'working with energy suppliers on interim commercial and technical solutions for increasing the likelihood of consumers keeping a smart service when they switch' in a self-governance mode.⁴³

3.6 The existing regulatory structure in the UK and Europe

Core regulatory bodies

The **Department for Business, Energy and Industrial Strategy (BEIS)**⁴⁴ holds the ministerial smart metering portfolio, previously held in the Department of Energy and Climate Change before a merger of departments in July 2016. On 30 March 2011, DECC (with Ofgem) published the Government's Response to the Smart Meter Prospectus. This set out a number of key dates relating to the rollout of smart meters in Great Britain. Since then, BEIS has continued to focus on ensuring all parties are making the necessary preparations for the main installation stage, so that energy suppliers are able to complete the rollout by the end of 2020.

A Smart Energy Code was put in place, necessary to enable the national data and communications system to go live⁴⁵. The Code details the rights and obligations of industry parties who use smart metering systems and the information they provide. According to BEIS, customers will be able to choose:

⁴² Pallas F. 2013 Beyond gut level - Some critical remarks on the German privacy approach to smart metering. *European Data Protection: Coming of Age* (Springer 2013). See http://link.springer.com/10.1007/978-94-007-5170-5_14 (accessed 26 June 2017).

⁴³ Written evidence submitted by Department of Energy and Climate Change (SME0024) to the House of Commons Science and Technology Committee. See <http://data.parliament.uk/writtenevidence/committeeevidence.svc/evidencedocument/science-and-technology-committee/smart-meters/written/34284.html> (accessed 26 June 2017).

⁴⁴ About Us - Department for Business, Energy & Industrial Strategy - GOV.UK. See <https://www.gov.uk/government/organisations/department-for-business-energy-and-industrial-strategy/about> (accessed 19 April 2017).

⁴⁵ SEC and Guidance Documents. See <https://www.smartenergycodecompany.co.uk/sec/sec-and-guidance-documents> (accessed 19 April 2017).

- how much data their energy supplier collects from their smart meter, e.g. monthly, daily or half-hourly meter readings;
- whether their supplier shares details about their energy consumption with other organisations;
- whether their supplier can use their meter reads for sales and marketing purposes;
- how they can access information about their energy use to get the most benefit from it.

The **Office of Gas and Electricity Markets (Ofgem)**⁴⁶ is a non-ministerial government department and an independent National Regulatory Authority. Its principal objective is to protect the interests of existing and future electricity and gas consumers. The **Gas and Electricity Markets Authority (GEMA)** serves as Ofgem's governing body overseeing Ofgem's work and providing strategic direction⁴⁷. The Authority's members are appointed by the Secretary of State at the Department for Business, Energy and Industrial Strategy.

Ofgem is providing independent regulatory expertise and advice to the government's central programme responsible for delivering the rollout of smart metering, taking steps to put in place appropriate consumer protections – including for customers who had smart meters installed before the completion of the government's regulatory framework for the smart meter roll-out. Ofgem has also taken on additional regulatory functions to support smart metering, including regulation of the new Data and Communications Company (DCC).

The Department of Business, Energy and Industrial Strategy (BEIS) held a competition to allocate a contract to a **Data and Communications Company (DCC)**⁴⁸ to establish and manage the data and communications network to connect smart meters to the business systems of energy suppliers, network operators and other authorised service users of the network. Smart DCC Ltd was granted a licence in September 2013, and is regulated by Ofgem. The data and communications infrastructure will:

1. operate consistently for all consumers regardless of their energy supplier;
2. provide smart metering data to network operators in support of smart grids;
3. allow authorised third parties to provide services to consumers who have granted them permission to use their data. Consumers can benefit by receiving energy services and advice on how to reduce their energy usage.

⁴⁶ About Us. 30 April 2013. See <https://www.ofgem.gov.uk/about-us> (accessed 19 April 2017).

⁴⁷ Transition to Smart Meters. 17 June 2013. See <https://www.ofgem.gov.uk/gas/retail-market/metering/transition-smart-meters> (accessed 19 April 2017).

⁴⁸ Data Communications Company. See <https://www.smartdcc.co.uk/about-dcc/> (accessed 19 April 2017).

European framework legislation⁴⁹:

Energy Efficiency Directive

The 2012 Energy Efficiency Directive⁵⁰ establishes a set of binding measures aimed at helping the EU to reach its 20% energy efficiency target by 2020. Under the Directive, all EU countries are required to use **energy** more efficiently at all stages of the energy chain, from production to final consumption.

On 30 November 2016, the Commission proposed an update to the Energy Efficiency Directive, including a new 30% energy efficiency target for 2030, and measures to update the Directive⁵¹ to make sure the new target is met. New national measures must seek to ensure major energy savings for consumers and industry alike. This includes ensuring that energy consumers should be empowered to better manage consumption. This includes easy and free access to data on consumption through individual metering.

Data Protection Act and GDPR

The Data Protection Act 1998 (and in future the GDPR) provides broad obligations that suppliers and other data controllers would need to meet. Following the developments outlined in the previous section, the Information Commissioner's Office (ICO) has supported the approach that the Government's policy on data access and privacy should address specific questions about the choices consumers should have, and the levels of energy consumption data that it is appropriate for suppliers and others to access to carry out essential functions connected to the provision of energy. The ICO also suggests that sector-specific provisions, that complement the Data Protection Act and the GDPR, might be appropriate in the case of smart metering.

Energy suppliers will need to comply with GDPR from May 2018. During the current transitional period, energy companies will need to work towards implementing GDPR provisions, including those relating to data portability, into their businesses. The supervisory authority under GDPR will be the Information Commissioner's Office, which is committed to working closely with other regulators, such as Ofgem.

It is the legal responsibility of all industry participants to ensure that they comply with the Data Protection Act and the GDPR to the extent that it applies to them. Generally speaking, suppliers, network operators and third parties accessing energy consumption data are likely to be data controllers, with the Data and Communications Company (DCC) potentially acting

⁴⁹ Relevant legislation include: Environmental Information Regulations 2004; The Gas Act 1986, the Electricity Act 1989; the Utilities Act 2000; the Competition Act 1998; the Enterprise Act 2002; the Energy Acts of 2004, 2008, 2010 and 2011; the Regulatory Enforcement and Sanctions Act 2008; Data Access and Privacy Framework (DAPF) 2012; Freedom of access to information Directive 2003; INSPIRE Regulations 2007; the Data Protection Act 1998; the General Data Protection Regulation, GDPR; Common Rules for the Internal Market in Electricity Directive (2009/72/EC); Common Rules for the Internal Market in Natural Gas Directive (2009/73/EC); Directive on the Processing of Personal Data (1995/46/EC); Energy Efficiency Directive (2012/27/EC); Regulation on guidelines for trans-European energy infrastructure ((EU) No 347/2013); Commission Recommendation of 9 March 2012 on preparations for the roll-out of smart metering systems (2012/148/EU); Commission Recommendation of 10 October 2014 on the Data Protection Impact Assessment Template for Smart Grid and Smart Metering; Directive 2014/94/EU on the deployment of alternative fuels infrastructure; Proposal for a revised electricity Directive.

⁵⁰ Energy Efficiency Directive - Energy - European Commission (*Energy*). See <https://ec.europa.eu/energy/en/topics/energy-efficiency/energy-efficiency-directive> (accessed 19 April 2017).

⁵¹ Commission Proposes New Rules for Consumer Centred Clean Energy Transition - Energy - European Commission (*Energy*). See <https://ec.europa.eu/energy/en/news/commission-proposes-new-rules-consumer-centred-clean-energy-transition> (accessed 19 April 2017).

as a data processor on their behalf, although this will depend on the exact nature of the activity being undertaken and the contractual basis for it⁵².

In addition to complying with the existing rules, the smart metering implementation programme considered privacy needs by adopting privacy by design principles throughout the smart metering regulatory regime. Under the GDPR, organisations have a general obligation to implement technical and organisational measures to show that they have considered and integrated data protection into their processing activities. Under the Data Protection Act, privacy by design has always been an implicit requirement of the principles - e.g. relevance and non-excessiveness. The programme recognised the confidentiality issues that may arise with respect to energy consumption data for non-domestic consumers and proposed that privacy by design principles should apply to both domestic and non-domestic consumers. According to Ofgem, in order to deliver “privacy by design” they will build on their initial work to carry out a detailed privacy impact assessment (PIA) and encouraging industry to carry out their own PIAs for their own internal systems⁵³.

Domestic energy regulation

The energy sector in the UK is mainly regulated through the Electricity Act 1989 and Gas Act 1986⁵⁴, both of which have been amended on numerous occasions to reflect developments in government policy. These Acts prohibit a number of activities, such as the supply of electricity, except under licence. Licence holders are required to comply with the relevant conditions contained within their licence. Compliance with these conditions is enforceable by Ofgem. Below these licence conditions sit a number of industry codes which contain the technical and commercial obligations that govern participation in licensed activities⁵⁵.

Significant changes to this regulatory framework have been made to achieve the government’s vision for every home to have smart metering equipment. These changes include:

1. amendments to existing energy licences and industry codes, for example to require suppliers to roll out smart meters, and consequential changes to legislation, licences and codes;
2. the introduction of a new licensable activity relating to communications between suppliers and other parties and smart meters in consumer premises and the appointment of a Data and Communications Company to carry out this licensed activity;
3. the introduction of a new Smart Energy Code. This sets out the rules, rights and obligations for all parties for the new enduring metering arrangements in Great Britain.

Government has been making the regulatory changes to implement these arrangements, principally using powers conferred on the Secretary of State by the Energy Act 2008 and

⁵² Smart Meter Data Access and Privacy - GOV.UK. See <https://www.gov.uk/government/consultations/smart-meter-data-access-and-privacy> (accessed 19 April 2017).

⁵³ Smart Meter Data Access and Privacy - GOV.UK. See <https://www.gov.uk/government/consultations/smart-meter-data-access-and-privacy> (accessed 19 April 2017).

⁵⁴ The Electricity Act 1989 provided for the privatisation of the electricity supply industry in Great Britain. The Act also established a licensing regime and a regulator for the industry called the Office of Electricity Regulation (OFFER), which has since become the Office of Gas and Electricity Markets (OFGEM). The Gas Act 1986 provided for the appointment and functions of a Director General of Gas Supplies and established the Gas Consumers Council. The Act also made new provisions for the supply of gas through pipelines.

⁵⁵ Smart Meters: Information for Industry and Other Stakeholders - GOV.UK. See <https://www.gov.uk/guidance/smart-meters-information-for-industry-and-other-stakeholders> (accessed 19 April 2017).

extended by the Energy Act 2011⁵⁶. Any changes made under these powers, must first be the subject of consultation, including with Ofgem, before being presented to Parliament⁵⁷. The changes to these licences and codes have been phased in tranches of regulation to give businesses and other stakeholders the time to give input on the detail of the regulatory framework through individual and detailed consultations.

The Orders

As part of the tranches of regulation changes BEIS used their powers under the Energy Act 2008 to make amendments to the Electricity Act 1989 and Gas Act 1986 (the 'Acts')⁵⁸, which enabled them to make two new statutory instruments that supported the establishment of the DCC. These are The Electricity and Gas (Smart Meters Licensable Activity) Order 2012 (SI 2012/2400)⁵⁹ and The Electricity and Gas (Competitive Tenders for Smart Meter Communication Licences) Regulations 2012 (SI 2012/2414) (DCC Licence Application Regulations)⁶⁰. The government also introduced a new statutory instrument to support the implementation of the Smart Energy Code (SEC). This is known as: The Electricity and Gas Appeals (Designation and Exclusion) Order 2013 (SI 2013/2429)⁶¹.

The Smart Energy Code (SEC)

When the Department of Energy and Climate Change (DECC) granted the Smart DCC Ltd licence, stage 1 of the Smart Energy Code (SEC) also came into force. The SEC is a new industry code and sets out the terms for the provision of the DCC's services and specifies other provisions to govern the end-to-end management of smart metering. Ofgem regulates DCC⁶² and like with other industry codes, they are responsible for approving any modifications to ensure consumers' interests remain protected.

The Smart Energy Code (SEC)⁶³ is a multi-party agreement, which requires energy suppliers, network operators and other relevant stakeholders to become a party to the SEC and to comply with its provisions in order to use DCC services. The SEC also sets out the rights and obligations of each Party. It is overseen by the SEC Panel and administered by the Smart Energy Code Administrator and Secretariat (SECAS). DCC is responsible for the development of a number of appendices known as subsidiary documents.

The Smart Energy Code Section 'Data Privacy' sets out the obligations regarding privacy, and are applicable to those DCC Users fulfilling the "Other User" role, as set out in the SEC. The Code acknowledges that, in providing the Services to a user, the DCC may act in the capacity of 'data processor' (as defined in the Data Protection Act 1998) on behalf of that user in respect of the Personal Data for which that user is the 'data controller' (as defined in the Data Protection Act), conferring on it duties set out in the DPA.

⁵⁶ Energy Act 2011. See <http://www.legislation.gov.uk/ukpga/2011/16/contents> (accessed 19 April 2017).

⁵⁷ Smart Meters: Information for Industry and Other Stakeholders - GOV.UK. See <https://paperpile.com/c/LDyx2u/TGVY> (accessed 19 April 2017).

⁵⁸ Ibid.

⁵⁹ The Electricity and Gas (Smart Meters Licensable Activity) Order 2012. See <http://www.legislation.gov.uk/ukdsi/2012/9780111526545/contents><http://www.legislation.gov.uk/ukdsi/2012/9780111526545/contents> (accessed 19 April 2017).

⁶⁰ The Electricity and Gas (Competitive Tenders for Smart Meter Communication Licences) Regulations 2012. See <http://www.legislation.gov.uk/ukdsi/2012/2414/introduction/made> (accessed 19 April 2017).

⁶¹ The Electricity and Gas Appeals (Designation and Exclusion) Order 2013. See <http://www.legislation.gov.uk/ukdsi/2013/2429/contents/made> (accessed 19 April 2017).

⁶² Transition to Smart Meters. 17 June 2013. See <https://www.ofgem.gov.uk/gas/retail-market/metering/transition-smart-meters> (accessed 19 April 2017).

⁶³ Smart Energy Code Home. See <https://www.smartenergycodecompany.co.uk/> (accessed 19 April 2017).

Privacy Assessments are undertaken to assess Parties' compliance with the obligations defined in the Code. The assessments are undertaken by an Independent Privacy Auditor, who has been appointed by the SEC Panel to provide the audit services. The Assessments must be carried out in accordance with a Privacy Controls Framework, which provides the basis for enabling a consistent level of review across all Users. The Framework includes:

- Arrangements designed to ensure that Privacy Assessments provide reasonable assurance that Other Users are complying with their obligations under Sections 1.2 to 1.5;
- The Principles and criteria to be applied in the carrying out of any Privacy Assessment, including principles designed to ensure that Privacy Assessments take place on a consistent basis across all Other Users; and
- The Provisions for determining the timing, frequency and selection of Other Users for the purposes of Random Sample Privacy Assessments.

3.7 Concluding remarks

The case of smart meters is multi-faceted and touches upon individual and systemic aspects of data governance. The UK experience bases its governance around a regulated trusted body, the DCC. The centrality of this body is only possible due to the central organisation and rollout of smart meter technology with the government lead, which sees the system architecture as purposefully constructed rather than emergent as a result of market forces. This creates a centralised morphology for data transmission and storage. An alternative structure, one based on cryptographic proofs providing privacy-respecting aggregation for network operators, as well as personalised billing was not taken up, with a number of possible reasons — lack of cryptographic expertise within government and business; the novelty of the application domain of smart meters; or concerns that privacy-enhancing measures might detract from the profitability of proposed systems.

Germany, another country with a smart metering target and a similar timeframe for its rollout, also opted against installing a data governance infrastructure based on novel privacy-enhancing technologies. However, the regulatory approach they took to metering infrastructure allowed the flexibility, through the specification of a new piece of intermediate hardware, to create new markets for privacy down the line.

Data governance is important in the context of increasingly, or even ubiquitously, smart infrastructure. Given the one-off nature of some systems there are no out-of-the-box solutions but will need researchers, businesses and government to work hand in hand to create regulation.